

Assel Kembay

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EDUCATION

University of California, Santa Cruz

Santa Cruz, CA, USA

Ph.D. in Electrical and Computer Engineering | GPA: 3.95/4.00

2028 (expected)

Area of Study: energy-efficient AI, brain-inspired ML | Advisor: *Prof. Jason Eshraghian*

Korea University of Science and Technology

Seoul, South Korea

M.S. in AI-Robotics | GPA: 4.43/4.50

Thesis: “Inversion of Spiking Neural Networks & its application to Knowledge Distillation”

C-DAC’s Advanced Computing Training School

Pune, India

Exchange student, postgraduate diploma in Advanced Computing

L.N. Gumilyov Eurasian National University

Astana, Kazakhstan

B.S. in Mathematical and Computer Modeling (summa cum laude equivalent) | GPA: 3.86/4.00

RESEARCH INTERESTS

Energy-efficient and neuromorphic AI systems; knowledge distillation and model compression; continual learning; brain-inspired machine learning.

AWARDS & HONORS

- Graduate Dean’s Research Travel Grant, UC Santa Cruz, USA 2026
- DAC 2025 Young Fellow, Design Automation Conference, San Francisco, USA (*competitive international selection*) 2025
- IEEE WIE Student Scholarship, International Leadership Conference, San Jose, USA 2025
- Graduate Studies DEI Research & Travel Award, UC Santa Cruz, USA 2025
- Divisional MIP Fellowship, UC Santa Cruz, USA (*merit-based, ~\$20K*) 2023
- POSCO Asia Fellowship, South Korea (*Next Generation Global Leaders program, full funding*) 2023
- KIST-KT&G Global Scholarship Foundation, South Korea 2021
- II Place, XV International Scientific Conference for Students and Young Scientists 2020
- Sur-Place Konrad Adenauer Foundation Scholarship, Germany 2019
- ITEC Programme Scholarship, Government of India (*full funding*) 2018
- Academic Achievement Award (seven times), ENU, Kazakhstan 2018
- Foundation of the First President of Kazakhstan Scholarship 2018

PUBLICATIONS

(* indicates equal contribution)

Full list: [Google Scholar](#) (166+ citations)

Journal Articles

- [J1] [APL Machine Learning 2026] **A. Kembay**, S. Gunasekaran, R.-J. Zhu, Y. Zhang, J. K. Eshraghian. (2026). *Efficient Knowledge Distillation via Salient Feature Masking*, APL Machine Learning 4, 016104 (2026). [\[paper\]](#) [\[code\]](#)
- [J2] [Nature Communications 2025] S. Gunasekaran, **A. Kembay**, et al. *Future-Guided Learning: A Predictive Approach to Enhance Time-Series Forecasting*, Nature Communications, 16(1), 8645. [\[paper\]](#) [\[code\]](#)

Conference and Workshop Papers

- [C1] [APCCAS 2026] **A. Kembay**, Jing Shuang (Lisa) Li, J. Eshraghian. *Complex-Pole Dynamics in Transformer Value Paths*, IEEE Asia Pacific Conference on Circuits and Systems (APCCAS), 2026.
- [C2] [IROS 2026] T. M. Huh, J. Lin, S. Cheng, **A. Kembay**, J. Eshraghian. *Sensing with Spikes: Neuromorphic Optical Tactile Sensor for Contact Force and Position Detection*, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.
- [C3] [WCCI IJCNN 2026] A. Smith, T. Kergan, **A. Kembay**, et al. *Training Spiking Neural Networks Using Lessons from State Space Models*, IEEE International Joint Conference on Neural Networks (IJCNN), 2026.
- [C4] [ISCAS 2025, DAC 2025] **A. Kembay***, K. Aguilar*, J. Eshraghian. *A Quantitative Analysis of Catastrophic Forgetting in Quantized Spiking Neural Networks*, 2025 IEEE International Symposium on Circuits and Systems (ISCAS). [\[paper\]](#) [\[poster\]](#) (also presented at DAC Young Fellows Poster Session)
- [C5] [IWCN 2021] Sch. Kim, Ch. Lee, B. Lee, D. Seol, D. Kim, **A. Kembay**, K. Yun, S. Jang, J. Lee. *Simulation Web Platform for the Electro-Chemical Oxygen Reduction Reaction*, 2021 International Workshop on Computational Nanotechnology (IWCN), **Oral**. (*Associated patent, 20% share*). [\[paper\]](#) [\[Project Page\]](#) [\[Platform Showcase\]](#)
- [C6] [KPS 2021] Sch. Kim, D. Kim, **A. Kembay**, S. Kim, K. Yun, et al. *Web Platforms for Conventional Simulations of Matters*, Korean Physical Society Spring Meeting (KPS), **Oral**. [\[paper\]](#) [\[video\]](#)

Preprints & Under Review

- [P1] Zhu R.-J.*, Peng T.*, ..., **Kembay A.**, ..., Eshraghian J. (2025). *A Survey on Latent Reasoning* (132 citations). [\[paper\]](#) [\[code\]](#)
- [P2] Y. Tian, **A. Kembay**, S. Tensingh, N. D. Truong, J. K. Eshraghian, O. Kavehei. (2025). *Beyond Pairwise Plasticity: Group-Level Spike Synchrony Facilitates Efficient Learning in Spiking Neural Networks*. [\[paper\]](#) [\[code\]](#)
- [P3] I. Aguilar, **A. Kembay**, J. K. Eshraghian, O. Kavehei. (2025). *A Neuroscience-Inspired Framework for Realistic Continual Learning: A Review*. [\[paper\]](#)

PATENTS

Electronic Structure Calculation Web-Program, Korea Institute of Science and Technology. **Inventors:** Kim Sch., **Kembay A.**, Kim S. (20% share each) [Project Page] [Platform Showcase]

RESEARCH EXPERIENCE

- **Research Associate, Conscium, London, UK** Mentor: [Dr. Max Ward](#)
Project: Generative modeling and optimization of mRNA sequences Remote | Jan. 2026 – Present
- **Research Scientist Intern, AI Fund, Mountain View, CA, USA** Mentor: [Dr. Andrew Ng](#)
Project: AI Systems for Vision-Based Reinforcement Learning in Interactive Environments Jan. 2026 – Apr. 2026
- **Graduate Student Researcher, University of California, Santa Cruz** Mentor: [Prof. Jason Eshraghian](#)
Project: Knowledge Distillation, Continual Learning and Efficient Language Models Oct. 2023 – Present
- **Research Scientist Intern, Korea University Medicine, Seoul, South Korea** Mentor: [Dr. Il-Joo Cho](#)
Project: Wireless Brain Chip Optimization and Data Transfer Algorithms Apr. 2023 – Sep. 2023
- **Research Assistant, AI Research Group, Korea Institute of Science and Technology** Mentor: [Dr. Suhyun Kim](#)
Project: Data-Free Knowledge Transfer for Neuromorphic Systems Sep. 2020 – Mar. 2023
- **Research Intern, Computational Science Research Center, KIST, Seoul, Korea** Mentor: [Dr. Seungchul Kim](#)
Project: Quantum Dot Simulation Platform Development Mar. 2020 – Aug. 2020

TEACHING EXPERIENCE

Teaching Assistant, ECE 173: High-Speed Digital Design (class size: 47), UC Santa Cruz Spring 2025

MENTORSHIP

Mentored 4 undergraduate researchers at UCSC, 2 of whom co-authored publications ([Nature Communications](#), [ISCAS](#)); and 15+ Kazakh/Central Asian students, who secured multi-year awards (Graduate Presidential Fellowship, DAAD) and fully-funded Ph.D. admissions (e.g., Columbia University).

Skye Gunasekaran (Undergraduate@UCSC) – [Nature Communications’25]	Dec. 2023 – Aug. 2025
Karina Aguilar (Undergraduate@UCSC) – [ISCAS’25]	May 2024 – Dec. 2024
Mariah Regalado (Undergraduate@UCSC)	Jun. 2025 – Oct. 2025
Yuchen Tian (Masters@University of Sydney, Australia)	Oct. 2024 – present

PROFESSIONAL SERVICE

Reviewer: Nature Communications ’26, NeurIPS NeuroAI ’24, APL Machine Learning ’24–26, IEEE ISCAS ’24–26, IEEE TCDS ’25–26, IEEE BioCAS ’26, ACM TOMM ’25

PROFESSIONAL MEMBERSHIPS

IEEE Graduate Student Member (Region 6 – Western USA, Monterey Bay Subsection); IEEE Women in Engineering (WIE); IEEE Circuits and Systems Society (CASS); Graduate Womxn in Computing, UCSC